Proceedings of the Fifth International Conference on Electronics, Communication and Aerospace Technology (ICECA 2021)
IEEE Xplore Part Number: CFP21J88-ART; ISBN: 978-1-6654-3524-6

Super-pixel Segmentation based Skin texture pattern recognition

Surya.T
Electrical and Electronics Engineering
Anna University-University College of Engineering
Thirukkuvalai, Tamil Nadu, India

suryasri.tm@gmail.com

Chitra Selvi .S
Electrical and Electronics Engineering
Anna University-University College of Engineering
Thirukkuvalai, Tamil Nadu, India

chitraselvieee@gmail.com

Arumugam Palanivel
Software Developer
KPIT Technologies Limited

35 & 36 Rajiv Gandhi Infotech Park, Hinjawadi, Pune, India
arumugam.elangovan@gmail.com

R.Bhuvaneswari,
Electronics and Communication Engineering
Indian Maritime University
Chennai, Tamil Nadu, India
bhuvanasraj@gmail.com

2021 5th International Conference on Electronics, Communication and Aerospace Technology (ICECA) | 978-1-6654-3524-6/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICECA52323.2021.9675925

Abstract- **Super-pixel segmentation is widely used nowadays in image processing to enhance segmentation accuracy. A new detection model is proposed for skin texture pattern recognition of Leopard, Cheetah, and Jaguar. In this model, a combination of Histogram of Gradient (HOG) and superpixel segmentation is used for extracting the features and the segmentation task of the target animal. This method does not require several superpixels to be created in advance, whereas it can automatically partition the image to its content into a suitable number of superpixels without any over or under segmentation. Then, the obtained features are fed into a Support Vector Machine (SVM) classifier to classify the skin texture pattern of Leopard, Cheetah, and Jaguar. The validation is performed which shows that the classifier achieves an accuracy of 96.67 %.**

Keywords— Super-pixel segmentation, Histogram of Gradient (HoG), Support Vector Machine (SVM)

I. INTRODUCTION

Nowadays automated computer vision application helps in identification of individual animal with its skin texture pattern. It's easy to differentiate different animals with their skin pattern. Biological studies and conservation efforts have made a manual recognition of individual animals with their shape and skin pattern. Texture and pattern play an important role in understanding and description of an object. Conservation research scientists deal with skin texture to recognize wild animals. It would be very helpful for the scientist with a system that can automatically identify animal skin texture patterns for the classification of wildlife with camera trap images. Some of the challenges in camera trap images are variations in point of view, occluded image, the dataset for testing, and training texture for classification. In this paper, samples of Leopard, Cheetah, and Jaguar are collected for skin texture pattern recognition. The skin texture pattern of Cheetah has a black solid spot throughout its body, Leopard has a rosette pattern and Jaguar have a rosette with a black solid spot in the center. Pattern recognition is the identification of patterns by employing a machine learning algorithm. It discriminates the input data based on the information gathered from the patterns of input data. Based on the recognition of skin pattern the classifier will give its decision whether it is Cheetah, Jaguar, or Leopard. There are many algorithms for pattern recognition such as Neural Network, Template matching, Fuzzy model, etc.

Super-pixel segmentation is implemented here for the segmentation of images because it holds more information compared to pixels. This gives a compact and convenient representation of an image. The goal of image segmentation is to provide an easier and meaningful representation of an image. For each pixel, a label is assigned in an image. Segmentation will locate the boundary and object in an image. Support Vector Machine (SVM) classifier is employed here for classification of skin texture pattern of Leopard, Cheetah, and Jaguar. SVM supports both sparse and dense sample vectors as input. A multi-class classification approach is implemented for skin texture pattern recognition of Leopard, Cheetah, and Jaguar. Hyperplanes in the SVM classifier helps in classifying the input data points. The dimensions of the hyperplane various with the number of features.

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

II. RELATED WORK

A new method of super-pixel segmentation using a distance function that is intended to balance boundary adherence, compactness characteristic, and intensity of resulting super-pixels. Here the author considered image boundaries to acquire super-pixels with effective boundary adherence. (i) Considered the edges of super-pixels that are persistent with object boundaries in an image. From this, a distance computation operation is applied here to restraint the homogeneity, boundary adherence, and compactness (COM). (ii) To relocate or to update the position and intensity of super-pixel seeds can be done by considering highly reliable pixels within the superpixel region [1]. An image segmentation using the superpixel method and the input image Initially segmented using local information-based simple linear iterative clustering (LI-SLIC). This can be executed with three steps: (i) Initial segmentation – the input image is segmented with LI-SLIC (ii) Hierarchical multilevel segmentation (iii) Merging superpixels [2]. A novel two-phase method for an accurate and automatic superpixel-based segmentation of breast ultrasound (BUS) images. In the first phase, to acquire an initial sketch of the tumor by disintegrating the BUS image into superpixel coarse to figure out their tumor likelihoods and by utilizing edge-based information which helps in quantifying the image region based information and contour cue that distinguishes the texture content of superpixels. In the next phase, the sketched tumor portion is enhanced by disintegrating the BUS image into superpixels with a large boundary recall ratio and the algorithm that segments superpixels occupied near the initial tumor outline [4]. An interactive segmentation by applying superpixels rather than the original pixels [5].

A novel denoising technique by integrating superpixel segmentation (SS) into low-rank representation (LRR) [6]. The Superpixel segmentation method is proposed to create comprehensive semantic information and structural superpixels along the sharp boundary [8]. An effective and robust method that enhances the identification of narrow and low contrast vessels. The author used superpixel as the fundamental unit of vessel segmentation [9]. Investigation of superpixel-based image representation is presented to obtain mid-level image features for image recognition to enhance accuracy [10]. Segmentation of synthetic aperture radar (SAR) image is proposed in [11].

An end-to-end trainable network to generate segmented and superpixel images simultaneously. Here a trained differentiable clustering algorithm is used to generate superpixels. Based on this, the superpixel pooling function is done to acquire superpixel features. Then similarity of two neighbor superpixels was calculated. If the similarity exceeds the threshold, the two neighbor superpixels were combined to acquire the segmented image [12]. An algorithm for automatic image segmentation with image-level labels and superpixels [13]. A method for identifying landslide deformation areas with video monitoring images [14]. Proposed a superpixel enhanced deep neural forest (SDNF) to overcome the challenges in semantic segmentation for a very high resolution (VHR) image. SDNF enhances the performance of the model from 91.7% to 92.6% [15].

TABLE 1 SUMMARY OF LITERATURE SURVEY

Reference Number	Year	Mertis	Limitations	Achieved accuracy
[3]	2018	A simple linear iterative clustering (SLIC) superpixel segmentation method and CNN-based deep learning technique is used for the classification of cellular structure in the histopathological image. The algorithm proposed can be enhanced by executing a more elaborated analysis of the m parameter and the superpixel number that affects each superpixel shape.	✓ To assist in the segmentation of multiple tumors in a single BUS image.	✓ The overall accuracy at classifying superpixels was achieved to be 0.9876. The run time of the CNN classification method was approximately 7750 seconds.
[7]	2012	Presented a multimodal semantic segmentation approach with the information of color and depth of an image.	✓ Other depth descriptors have to be evaluated with many complex databases.	✓ Obtained recognition rate around 80%

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

[22]	2021	The author proposed an early identification of the Hard execute (HE) pointed in the blood vessels of an eye due to diabetic condition by implementing dense deep convolutional neural network (CNN) architecture.	✓ The architecture has to be extended to incorporate images from the unfiltered world.	✓ Achieved 97 % Accuracy and 89.5 % Precision
[23]	2021	The author proposed fusion-based feature extraction for the interpretation of ECG signals. The signal can be affected by various noises such as Instrumentation, calibration, and thermal. These distractions can be removed by the de-noising filter. The de-noising filter output is the input for fusion-based extraction of features and construction of the decision model.	✓ The classification of large irrelevant data is one of the challenging tasks in this algorithm. As well a the accuracy and computation time can be improved by removing features of the correlated function.	✓ The accuracy achieved using fusion technique architecture with 350 samples. Accuracy= 96.5%, Sensitivity= 94.35%, Specificity= 97.20%.
[24]	2021	The author suggested an extreme learning machine (ELM) for various classification tasks. This ELM is the re-designed model of the neural network with its components such as weights, hidden layers, and biases. These variants of ELM utilizes methods adjusting based on the hidden nodes in the neural structural layer. This king of pruning ELM improves the learning time and verifies the active node number in the layer.	✓ ELM can be implemented for pattern recognition, medical image diagnosis, etc in a single architecture framework with the distributed computation methods.	✓ The accuracy achieved is 94.10 %. ✓ The execution time taken is 58 seconds.

III. DATASET

The image dataset of three animals namely Leopard, Cheetah, and Jaguar are considered for their skin texture recognition. The Leopard image dataset was collected from the forest officials and the images of Cheetah and Jaguar are manually downloaded from different websites. 500 images of these 3 animals are considered in different poses. Out of 500 images, 400 were used for validation and training and 100 images were used for testing.

Fig. 1. Dataset images of Leopard, Cheetah, and Jaguar

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

IV. PROCESSING OF SKIN TEXTURE PATTERN RECOGNITION OF LEOPARD, CHEETAH, AND JAGUAR

A. Pre-processing

The main purpose for performing pre-processing the input image is to enrich some image features before future processing and to improve the image data that are suppressed with some unwanted distortion. It mainly aims to stabilize the intensity of the image. It can be carried out by image resizing and RGB to Grayscale conversion.

- *RGB to Grayscale conversion*

RGB color image needs 8 bits for each color component ie. 8*3=24 bits are needed to keep a single color pixel of an image. When RGB is converted to Gray, it needs only 8 bits to store a single pixel of an image. 33% less memory is needed to keep a grayscale image than an RGB image. Grayscale images are very easier to workout many tasks like image segmentation, Morphological operation, etc., because it's a single-layered image 1D (Gray Scale image) not like a three-layered image 3D (RGB color image). Grayscale conversion changes the brightness of the image without regard to the position in the image. It does not depend on the position of the pixel in the image. It's easier to differentiate aspects of an image when we handle the single-layered image (Grayscale image). RGB to Grayscale conversion can be done with two methods namely weighted or luminosity method and average method. Here, it is performed with the average method. An average method is taking the average of three colors. It means taking r,g,b then divide it by 3 to get the defined grayscale image.

$$\text{Grayscale image} = \frac{R+G+B}{3} \quad (1)$$

- *Image Resizing*

Image Resizing is essential when it is needed to expand or shrink the number of pixels in an image. Resizing can be done through image compression which can diminish the size of the image, but the image quality gets to suffer and may lose some data in an image. Many digital cameras can perform a digital zoom. The details are not clear even though the photo contains the same number of pixels with digital zoom. Zooming increases the number of pixels in an image so that it is possible to see more details when an image is zoomed. All the image samples are kept in 227*227*3 pixels.

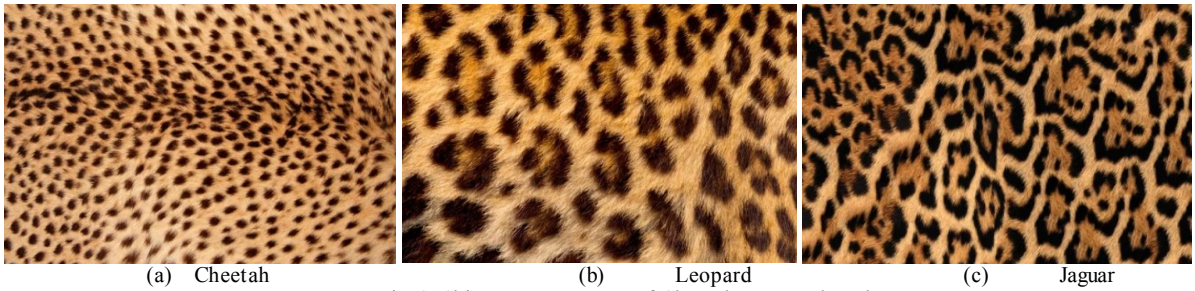
B. Super-Pixel Segmentation

Super-pixel means a cluster of connected pixels with the same colors or gray levels. Superpixel segmentation is partitioning the image into hundreds of non-overlapping superpixels. Super-pixel segmentation is applied for certain computer vision like image categorization, image segmentation, visual tracking, etc., Here super-pixel segmentation for classifying Cheetah, Leopard, and Jaguar based on their skin texture pattern. Cheetah has a black spot throughout its body, Leopard has a rosette pattern and Jaguar have a black spot at the center of the rosette pattern as shown in Fig. 2 (a,b,c). An image is concealed with several overlapping square patches are of definite size and each pixel is encountered by certain patches and the task is to allocate a pixel to one of the patches. If two adjacent pixels are allocated to the same patch, there is no penalty. If two adjacent pixels are assigned to various patches, then there is a stitching penalty that is inversely proportional to the intensity variations among the pixels. Extraction of superpixels is widely employed for classification of images and annotations [28].

C. Histogram of Oriented Gradient (HoG)

In the HOG feature descriptor, the histogram is the distribution, of Oriented Gradient represent the direction of gradient are used as features and the gradients- the magnitude of the gradient is high around corners and edges. HOG is used for object detection in an image. HOG is utilized as a feature descriptor that is used to extract image features from the input image. A feature descriptor is an image patch that simplifies the image by extracting useful information in an image and neglect other extraneous information. The size of the image is converted into width x height x 3 (channels) to an array of length n to a feature vector. In the HOG feature descriptor, the size of an input image is 64*128*3 and the output length of the feature vector is 3780. Feature vectors are not only used for

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.



(a) Cheetah (b) Leopard (c) Jaguar

Fig. 2. Skin texture pattern of Cheetah, Leopard, and Jaguar

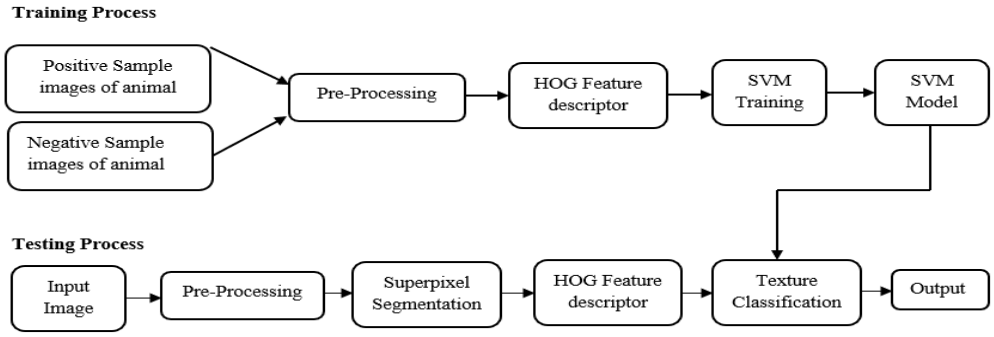


Fig. 3. Block Diagram for Skin texture pattern recognition of Leopard, Cheetah, and Jaguar

viewing the image it is also applied for certain tasks like object detection and image recognition. The feature vector made by these algorithms when applied to an image categorization algorithm support vector machine (SVM) achieves an excellent result.

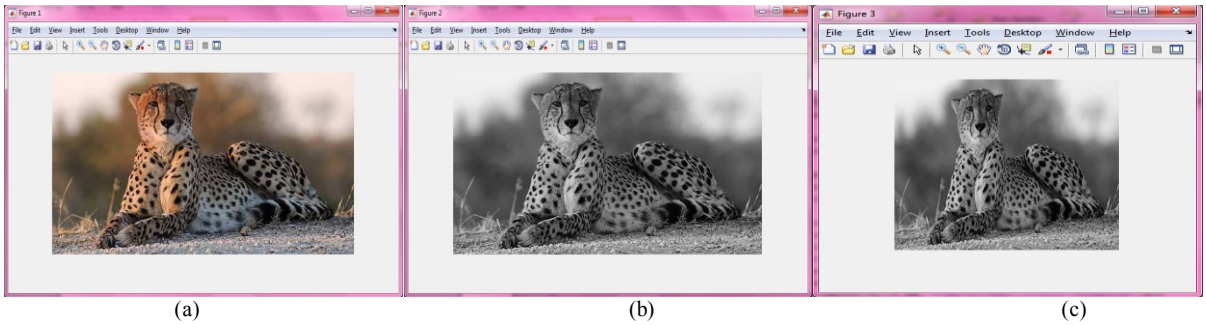
D. Support Vector Machine (SVM) Classification process

SVM is the most robust and accurate classification algorithm for classifying the skin texture pattern of the Cheetah, Leopard, and Jaguar. SVM is the method of statistical learning that works based on interval maximization that plots the inseparable linear feature vectors into high dimensional features. So, that the feature vectors are separable linearly in high dimensional features. It achieves an excellent classification accuracy with the finite number of training samples because of its finite generalization ability [16]. To segregate nonlinear separable information that is mapped onto space with a higher dimension. It needs more complex computations, which functions very slow. Hence, kernel functions are utilized for the segregation of non-linear information linearly based on this conversion, a perfect boundary can be drawn for portioning and this will not be a straight line. The main benefit of using kernel functions is it generates a connection among the data points and reduces complex computation [17]. The author trained three-site distinct SVM models, to distinguish between water and surface ice and generated binary images from whole surface ice concentration [18].

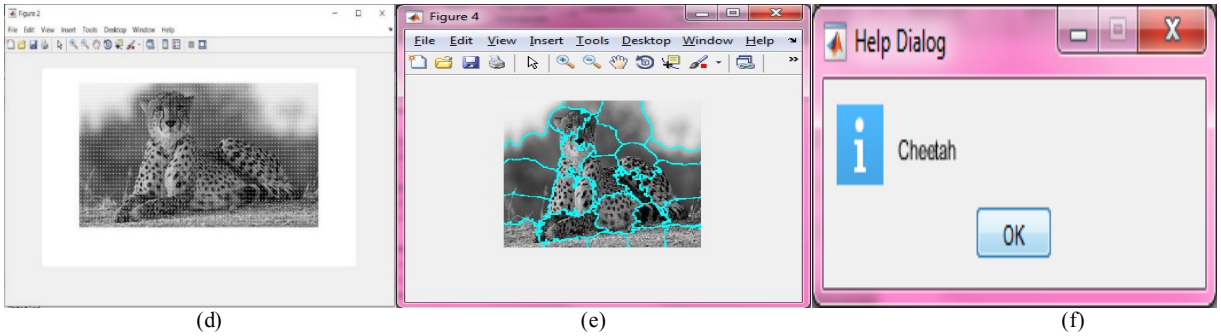
An experiment for classification among two locust species based on the SVM classification algorithm and attained a semi-automated accuracy of 96.16% [19]. The author investigated the Deep Support vector machine (DSVM) carried out with four kernel functions for the classification of hyperspectral images [20]. An innovative method for the segmentation of medical images by assimilating graph cuts and support vector machine (SVM) [21]. SVM is a widely used algorithm for pattern recognition for the reason of its ability to segregate unseen data and good analyzing performance. Many empirical risk techniques such as decision trees, Artificial neural networks (ANN), etc can also be effective but SVM is superior to those techniques as well as different variants are developed for performance enhancement [25]. The author proposed a SVM multi class classifier for facial recognition and also compared their model result with other classification model such as convolutional neural network (CNN), Hidden Markov Model (HMM), Nearest feature line (NFL). SVM outperforms with 3 % rate of error on face dataset [26]. The author presented the detected viral pattern of COVID-19 for this the author utilized Histogram Oriented Gradient (HOG), thoracic Xrays for extraction of features. Further, the author proposed a CNN classification model for medical imaging that can be utilized based on various edge-based neural networks. This CNN method of classification is tested and trained with a public X-ray database [27].

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

The results for skin texture pattern recognition of Cheetah, Leopard, and Jaguar were taken in MATLAB 2019b environment.

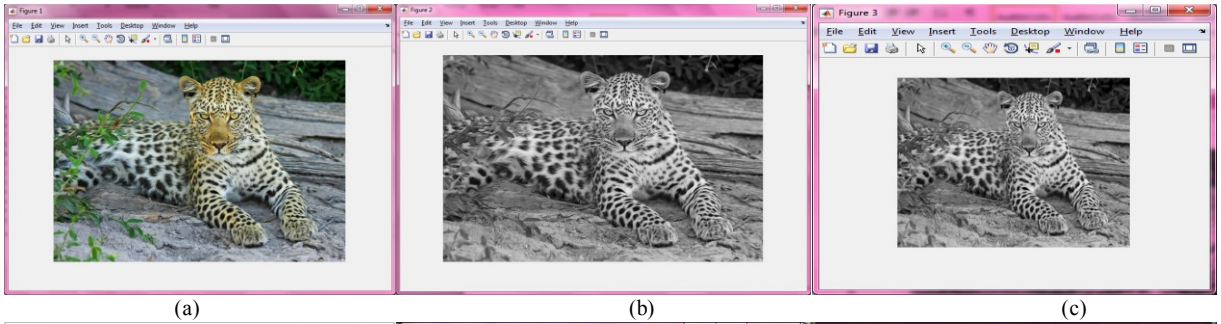


(a) (b) (c)

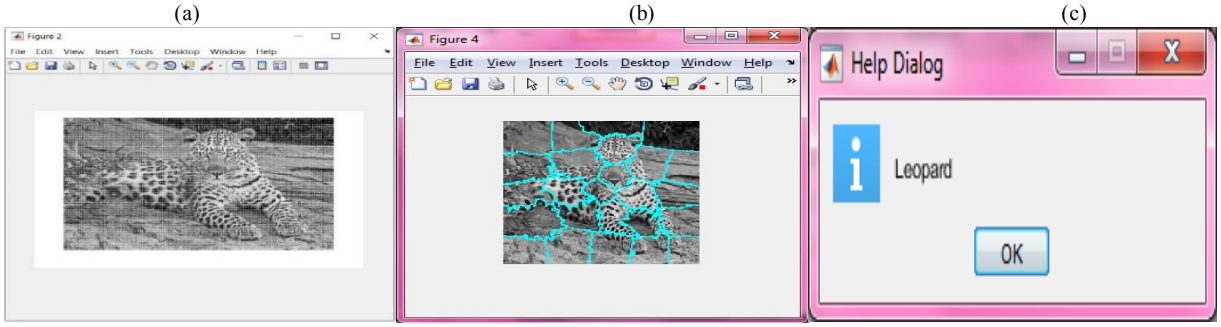


(d) (e) (f)

Fig. 4. Result for cheetah skin texture pattern (a) Input (b) RGB to grayscale conversion (c) Image resizing (d) HOG (e) superpixel segmentation (f) SVM decision



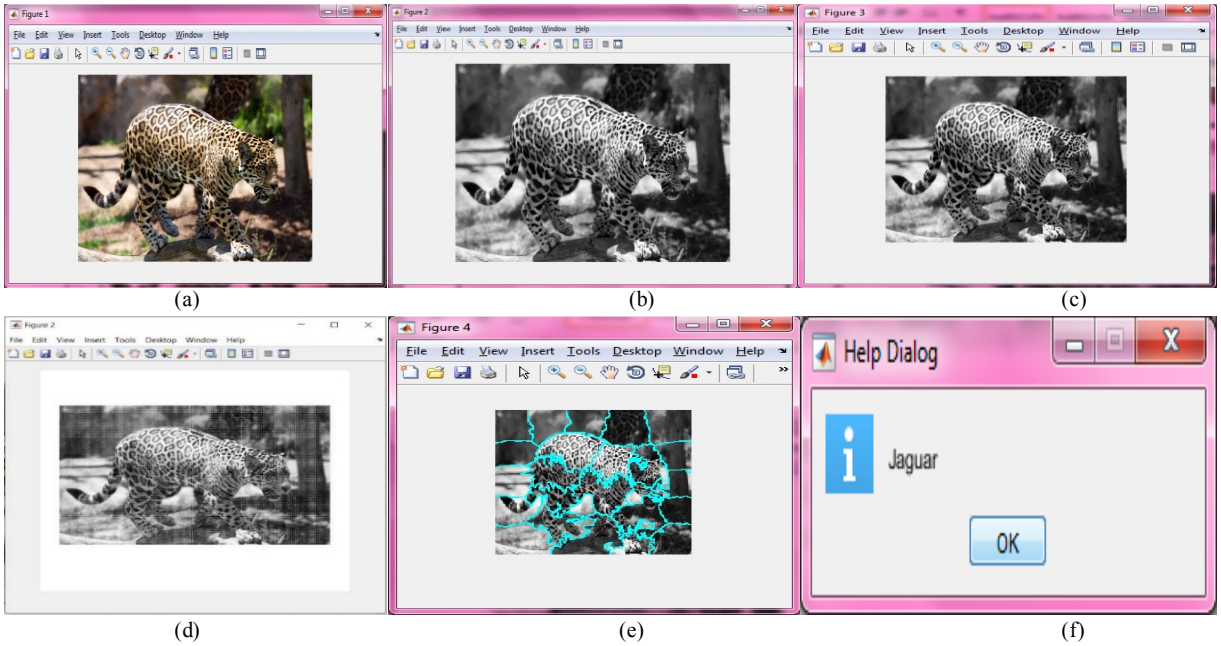
(a) (b) (c)



(d) (e) (f)

Fig. 5. Result for Leopard skin texture pattern (a) Input (b) RGB to grayscale conversion (c) Image resizing (d) HOG (e) Superpixel segmentation (f) SVM decision

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

(d) (e) (f)

Fig. 6. Result for Jaguar skin texture pattern (a) Input (b) RGB to grayscale conversion (c) Image resizing (d) HOG (e) superpixel segmentation (f) SVM decision

V. VALIDATION

The dataset for this experiment was gathered from online resources and the forest department. During the training phase, almost 80 image samples for each animal (Cheetah, Leopard, and Jaguar) are utilized, for the testing phase, 50 image samples are taken for each animal, The performance is calculated depending on the total cross-validation.

$$Accuracy = \frac{\text{True positive (TP)}}{\text{True positive (TP)} + \text{False Negative (FN)}} \quad (2)$$

$$Precision = \frac{\text{True positive (TP)}}{\text{True positive (TP)} + \text{False Positive (FP)}} \quad (3)$$

True Positive (TP) - Correctly recognized

False Positive (FP) - Wrongly recognized

False Negative (FN) - Failed to recognize

Performance validation shows that the Support Vector Machine (SVM) algorithm attains good performance in the classification of skin texture pattern recognition of Cheetah, Leopard, and Jaguar as shown in Table 2. The performance of the various classifiers for skin texture pattern recognition is shown in Table 3.

TABLE II PERFORMANCE EVALUATION OF SUPPORT VECTOR MACHINE (SVM) FOR SKIN TEXTURE PATTERN RECOGNITION OF CHEETAH, LEOPARD, AND JAGUAR

Samples	Training Samples	Testing Samples	TP	FP	FN	Accuracy (%)	Precision (%)
Cheetah	80	50	49	2	1	98	96.08
Leopard	80	50	48	2	2	96	96
Jaguar	80	50	47	1	3	94	97.91

TABLE III PERFORMANCE EVALUATION OF DIFFERENT CLASSIFIERS FOR SKIN TEXTURE PATTERN RECOGNITION OF CHEETAH, LEOPARD, AND JAGUAR

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

Type of classifier	Training Time(s)	Prediction time (s)	Accuracy (%)	Precision (%)
Adaboost	120	2.0	91.56	90.89
Neural Network	115	1.2	93.27	92.14
SVM (Proposed Classifier)	112	1.0	96	96.67

VI. CONCLUSION

In this work, a novel scheme is proposed for skin texture pattern recognition of the Cheetah, Leopard, and Jaguar. The combination of Histogram of gradient (HOG) and superpixel segmentation which is mainly used for feature extraction and segmentation task and Support Vector Machine (SVM) model is performed for the discrimination of skin pattern of Cheetah, Leopard, and Jaguar. The SVM classifier shows outstanding performance with the limited number of the dataset.

ACKNOWLEDGMENT

The authors are highly grateful to the Supervisor Dr. S. Chitra Selvi. I sincerely acknowledge the University Grants Commission (UGC), New Delhi, India for providing the fellowship through Rajiv Gandhi National Fellowship (RGNF) Scheme.

REFERENCES

[1] Y. Zhang, X. Li, X. Gao and C. Zhang, "A Simple Algorithm of Superpixel Segmentation With Boundary Constraint," in IEEE Transactions on Circuits and Systems for Video Technology, vol. 27, no. 7, pp. 1502-1514, July 2017, DOI: 10.1109/TCSVT.2016.2539839.

[2] Shuan-hu, Di, M. Liao, Yu-qian Zhao, L. Yang and Yezhan Zeng. "Image superpixel segmentation based on hierarchical multi-level LI-SLIC." *Optics and Laser Technology* 135 (2021): 106703.

[3] A. Albayrak and G. Bilgin, "A Hybrid Method of Superpixel Segmentation Algorithm and Deep Learning Method in Histopathological Image Segmentation," 2018 Innovations in Intelligent Systems and Applications (INISTA), Thessaloniki, Greece, 2018, pp. 1-5, DOI: 10.1109/INISTA.2018.8466281.

[4] Daoud, Mohammad & Atallah, Ayman & Awwad, Falah & Al najar, Mahasen & Alazrai, Rami. (2018). Automatic Superpixel-based Segmentation Method for Breast Ultrasound Images. Expert Systems with Applications. 121. doi:10.1016/j.eswa.2018.11.024.

[5] J. Ding, C. Lin, I. Lu and Y. Cheng, "Real-time interactive image segmentation using improved superpixels," 2015 IEEE International Conference on Digital Signal Processing (DSP), Singapore, 2015, pp. 740-744, DOI: 10.1109/ICDSP.2015.7251974.

[6] Fan Fan, Yong Ma, Chang Li, Xiaoguang Mei, Jun Huang, Jiayi Ma, Hyper-spectral Image Denoising with Superpixel Segmentation and Low-Rank Representation, Information Sciences (2017), DOI: 10.1016/j.ins.2017.02.044

[7] Jebari, Islem & Filliat, David. (2012). Color and Depth-Based Superpixels for Background and Object Segmentation. Procedia Engineering. 41. 1307-1315. 10.1016/j.proeng.2012.07.315.

[8] H. Li, S. Kwong, C. Chen, Y. Jia and R. Cong, "Superpixel Segmentation Based on Square-Wise Asymmetric Partition and Structural Approximation," in IEEE Transactions on Multimedia, vol. 21, no. 10, pp. 2625-2637. Oct. 2019, DOI: 10.1109/TMM.2019.2907047.

[9] B. Sheng *et al.*, "Retinal Vessel Segmentation Using Minimum Spanning Superpixel Tree Detector," in *IEEE Transactions on Cybernetics*, vol. 49, no. 7, pp. 2707-2719, July 2019, DOI: 10.1109/TCYB.2018.2833963.

[10] H. Emrah Tasli, Ronan Sicre, and Theo Gevers. 2015. SuperPixel based mid-level image description for image recognition. J. Vis. Comun. Image Represent. 33, C (November 2015), 301–308. DOI:https://doi.org/10.1016/j.jvcir.2015.09.021.

[11] X. Tian, L. Jiao, L. Yi, K. Guo, X. Zhang, The image segmentation based on optimized spatial feature of superpixel, J. Vis. Commun. Image R. (2014), DOI: http://dx.doi.org/10.1016/j.jvcir.2014.11.005.

[12] Kai Wang, Liang Li, Jiawan Zhang, End-to-end trainable network for superpixel and image segmentation, Pattern Recognition Letters (2020), DOI: https://doi.org/10.1016/j.patrec.2020.09.016.

[13] X. Xie, G. Xie, X. Xu, L. Cui and J. Ren, "Automatic Image Segmentation With Superpixels and Image-Level Labels." in *IEEE Access*, vol. 7, pp. 10999-11009, 2019, DOI: 10.1109/ACCESS.2019.2891941.

[14] Yang, Yang & Song, Shuliang & Yue, Fucai & He, Wen & Shao, Wei & Zhao, Kui & Nie, Wen. (2019). Superpixel-based automatic image recognition for landslide deformation areas. Engineering Geology. 259. 105166. 10.1016/j.enggeo.2019.105166.

[15] Mi, Li and Z. Chen. "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation." Isprs Journal of Photogrammetry and Remote Sensing 159 (2020): 140-152.

[16] Baoshi Chen , Lingling Zhang , Hongyan Chen , Kewei Liang , Xuzhu Chen , A Novel Extended Kalman Filter with Support Vector Machine Based Method for the Automatic Diagnosis and Segmentation of Brain Tumors, Computer Methods and Programs in Biomedicine (2020), DOI: https://doi.org/10.1016/j.cmpb.2020.105797

[17] Azarmdel, Hossein & Jahanbakhshi, Ahmad & Mohtasebi, Seyed & Rosado, Alfredo. (2020). Evaluation of image processing technique as an expert system in mulberry fruit grading based on ripeness level using artificial neural networks (ANNs) and support vector machine (SVM). Postharvest Biology and Technology. 166. DOI: 111201. 10.1016/j.postharvbio.2020.111201.

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.

[18] H. Kalke, M. Loewen , Support vector machine learning applied to digital images of river ice conditions. Coltec (2018), DOI :10.1016/ j.coldregions.2018.08.014.

[19] LU, Shuhan & Ye, Sijing. (2020). Using an image segmentation and support vector machine method for identifying two locust species and instars. Journal of Integrative Agriculture. 19. 1301-1313. DOI: 10.1016/S2095-3119(19)62865-0.

[20] Okwuashi, Onuwa & Ndehedehe, Christopher. (2020). Deep support vector machine for hyperspectral image classification. Pattern Recognition. 103. 107298. 10.1016/j.patcog.2020.107298.

[21] Q. Zheng, H. Li, B. Fan, S. Wu, J. Xu, Integrating support vector machine and graph cuts for medical image segmentation, J. Vis. Commun. Image R. (2018), DOI: https://doi.org/10.1016/j.jvcir.2018.06.005.

[22] Sungheetha, A. & R, R. S. "Design an Early Detection and Classification for Diabetic Retinopathy by Deep Feature Extraction based Convolution Neural Network", Journal of Trends in Computer Science and Smart Technology, 3(2), 81-94, 2021. doi:10.36548/jtcsst.2021.2.002

[23] Vijayakumar, D. T., Vinothkanna, M. R. & Duraipandian, D. M. "Fusion based Feature Extraction Analysis of ECG Signal Interpretation - A Systematic Approach". Journal of Artificial Intelligence and Capsule Networks, 3(1), 1-16, 2021. doi:10.36548/jaicn.2021.1.001.

[24] Manoharan, D. J. S. "Study of Variants of Extreme Learning Machine (ELM) Brands and its Performance Measure on Classification Algorithm". Journal of Soft Computing Paradigm. 3(2), 83-95, 2021.doi:10.36548/jscp.2021.2.003.

[25] Borah, P., & Gupta, D. (2017). Review: Support Vector Machines in Pattern Recognition. International Journal of Engineering and Technology. 9(3S), 43–48. doi:10.21817/ijet/2017/v9i3/17090

[26] Guo, Guodong, Stan Z. Li, and Kapluk Chan. "Face recognition by support vector machines." Automatic Face and Gesture Recognition, 2000. Proceedings. Fourth IEEE International Conference on. IEEE, 2000.

[27] Chen, Joy. (2021). Design of Accurate Classification of COVID-19 Disease in X-Ray Images Using Deep Learning Approach. Journal of ISMAC. 2. 132-148. 10.36548/jismac.2021.2.006.

[28] H. Sima and P. Guo, "Texture Superpixels Merging by Color-texture Histograms for Color Image Segmentation," KSII Transactions on Internet and Information Systems. vol. 8. no. 7, pp. 2400-2419, 2014. DOI: 10.3837/tiis.2014.07.011.

Authorized licensed use limited to: Jain University. Downloaded on April 21,2026 at 04:45:38 UTC from IEEE Xplore. Restrictions apply.